

Open Source Technologies

(01CE0526)

Lab Manual 2026-27

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Class- 5EV7

Rubrics for Lab

Component	Marks	Assessment Criteria
Problem Understanding & Approach	3Marks	Understanding of the experiment, selection and usage of Git, GitHub, Laravel, MongoDB, PostgreSQL, Selenium, Postman, JMeter, or other relevant open-source tools.
Implementation & Execution	4Marks	Correct execution of commands, coding/configuration, workflow implementation, successful completion of the practical, and error handling.
Output Analysis & Viva	3Marks	Correct output verification, interpretation of results, proper documentation, and ability to answer viva questions.

Sr. No	Program	Co	Date	R1	R2	R3	Marks	Signature
1	Explore GitHub/GitLab for open-source projects with different licenses.	CO1						
2	Setup git and explore commands related to VCS.	CO2						
3	Create a GitHub/GitLab repository and upload sample code.	CO2						
4	Implement CI/CD Pipeline using GitHub Actions / GitLab CI.	CO2						
5	Develop a Laravel/Django based web application.	CO3						
6	Deploy a simple app using Nginx/Apache.	CO3						
7	Setup Nginx to handle proxy request and load balancing.	CO3						
8	Use MongoDB to create a NoSQL database, perform CRUD operations, and query documents using basic operators.	CO4						
9	Use PostgreSQL to create a relational database, design tables, and perform CRUD operations using SQL queries.	CO4						
10	Use pytest to test python application.	CO5						
11	Use selenium to create browser-based test.	CO5						
12	Use Postman to test sample API.	CO5						
13	Use OWASP ZAP to check security integrity.	CO5						
14	Use JMeter to test web application for load testing.	CO5						

Note:

1. All Students Will perform Practical as per the Given Lab Manual Only.
2. All Students need to use this Index only.
3. For Lab File: Mandatory Follow OST Subject Lab Print Format. All the Students need to take Hard Copy of Practical in Given Format only.
4. LAB#1 to LAB#14 Submission and Assessment will be done in both, file and Google Class Room.

Prof. Priyanka Mangi
Subject Faculty

Dr. Krunal Vaghela
Associate Dean & Head of the Department

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- Content Font size : Calibri 12
- Screenshot heading Font Size : 9 Calibri

Practical 1 : Explore GitHub/GitLab for open-source projects with different licenses.

Note : Explore 10 different projects with their licenses

Sr. No.	Repository	Platform	Purpose	License	Can Modify	Commercial Use	Must Share Changes
1	React	GitHub	JavaScript Library	MIT	Yes	Yes	No
2.	Hyprrland	GitHub	Wayland Compositor	BSD 3-Clause	Yes	Yes	No
3.	Immich	GitHub	Self-hosted Photo and Video Backup	AGPL	Yes	Yes	Yes
4.	FFmpeg	GitHub	Multimedia processing library	LGPL	Yes	Yes	No
5.	Linux	GitHub	Operating System Kernel	GPL-2.0	Yes	Yes	Yes
6.	Ani-cli	GitHub	Anime Streaming CLI	GPL-3.0	Yes	Yes	Yes
7.	Odysseus	GitHub	Self-hosted AI workspace	AGPL-3.0	Yes	Yes	Yes
8.	Resonance	GitHub	Self-hosted Music Player	MIT	Yes	Yes	No
9.	Firefox	GitHub	Web Browser	MPL-2.0	Yes	Yes	No
10.	Ollama	GitHub	Run LLMs Locally	MIT	Yes	Yes	No

Practical 1: Explore GitHub/GitLab for open-source projects with different licenses.

Aim:

To install Git, create a GitHub repository, and explore basic Version Control System (VCS) operations.

1: Creating a New Repository

The screenshot shows the GitHub 'New repository' page. The 'General' tab is selected, displaying the repository name 'open_source_technology' and the description 'practical-1'. The 'Configuration' tab is also visible, showing the visibility set to 'Public' and options to add a README, .gitignore, and license, all of which are currently set to 'No'.

2: Creating and committing the initial File

The screenshot shows the GitHub repository page for 'open_source_technology'. The repository is public and shows a commit history with one commit 'Create index.html'. The README section is empty, with a prompt to 'Add a README'. The right sidebar shows repository statistics: 0 stars, 0 watching, and 0 forks.

3: Editing index.html file

The screenshot shows a web browser window with the GitHub repository editor interface. The repository is named 'open_source_technology' and the file being edited is 'index.html'. The editor shows the following content:

```
1 Hello ,
2 this is raviteja
```

The interface includes a sidebar with a file explorer showing 'index.html' and a main editing area with 'Edit' and 'Preview' tabs. The 'Edit' tab is active. The bottom of the browser shows a Windows taskbar with various application icons and a system clock indicating 9:08 AM on 7/20/2026.

4: Exploring a public GitHub Repository

The screenshot shows the GitHub repository page for 'Angular-Full-Stack' by user 'DavideViolante'. The repository is public and has 0 stars, 0 forks, and 0 watches. The description states: 'Angular Full Stack project built using Angular, Express, Mongoose and Node. Whole stack in TypeScript.' The repository is forked from 'DavideViolante/Angular-Full-Stack'.

The repository has 1 branch (master) and 0 tags. The commit history shows the following commits:

Commit	Message	Author	Date
40e63f4	chore: add contributors section to README [skip ci]	DavideViolante	5 months ago
	feat: use newest Angular features such as signals and more +...		5 months ago
	chore(deps): add husky and pre-commit hook to lint and build		2 years ago
	feat: use newest Angular features such as signals and more +...		5 months ago
	chore(deps): migrate to express v5		5 months ago
	Add .dockerignore		7 years ago
	feat: use newest Angular features such as signals and more +...		5 months ago
	Server side testing (DavideViolante#104)		9 years ago
	feat: use newest Angular features such as signals and more +...		5 months ago
	Add html and ece link (DavideViolante#105)		8 years ago

The repository also includes a sidebar with links to Readme, MIT license, Activity, 0 stars, 0 watching, and 0 forks. The 'Releases' section shows 'No releases published' and a link to 'Create a new release'. The 'Packages' section shows 'No packages published' and a link to 'Publish your first package'.

5: Repository import completed successfully

https://github.com/raviteja0629/react-copy/import

aA ☆ ⚙️ |  Chat

raviteja0629 / react-copy

🔍 Type to search

     

<> Code

🔍 Issues

↕️ Pull requests

🔄 Actions

📁 Projects

🛡️ Security and quality

📊 Insights

⚙️ Settings

Preparing your new repository

There is no need to keep this window open. We'll email you when the import is done.



Importing complete! Your new repository [raviteja0629/react-copy](#) is ready.